

Data Encryption

Learning Objectives

- Understand why data needs protection
- Explain the concept of encryption and decryption
- Identify everyday uses of encryption

Engage (5-7 minutes)

Think of sending a secret letter written in a code only your friend can read. Even if someone intercepts it, they cannot understand it. Data encryption works the same way—it scrambles data so only authorized people can read it.

Explore (10-12 minutes)

Try a simple substitution cipher: replace each letter with one three positions ahead (A becomes D, B becomes E). Write a short encrypted message and exchange with a partner. Try to decode each other's messages. Discuss what makes a code strong.

Explain (10-12 minutes)

Encryption converts readable data (plaintext) into scrambled form (ciphertext) using a key. Decryption reverses it with the same or related key. HTTPS on websites, UPI payments, and WhatsApp messages all use encryption to protect data from hackers during transmission.

Elaborate (8-10 minutes)

When you check bank balance on phone, encryption protects your PIN and account number. List five daily activities that use encryption: online shopping, email, messaging, logging into websites, digital payments. Explain what could happen without encryption.

Evaluate (5-8 minutes)

Exit ticket: Define encryption and decryption. Name two everyday applications that use encryption to protect data.